UNION BOUNDS FOR THE POINCARÉ CONSTANT

SAM POWER

Abstract. The Poincaré constant of a probability measure quantifies the connectedness of the domain relative to the distribution of mass. A textbook result states that given a probability measure and bounds on its Poincaré constant on two overlapping sets, one can deduce a bound on its Poincaré constant on the union of those two sets. The textbook bound can be sharpened somewhat, and the present note details how to do so.

Notation and Setup

Let μ be a probability measure on $\mathcal{X} \subseteq \mathbf{R}^d$. For any measurable set $S \subseteq \mathbf{R}^d$ with $\mu(S) > 0$, we define the mean, unnormalized variance and the Dirichlet energy of a sufficiently regular function $f : \mathcal{X} \to \mathbf{R}$ by

$$\begin{aligned} m_\mu(f;S) &:= \mu(S)^{-1} \int_S f \, \mathrm{d}\mu \\ \mathrm{Var}_\mu(f;S) &:= \inf_{t \in \mathbf{R}} \int_S |f-t|^2 \, \mathrm{d}\mu = \int_S |f - m_\mu(f;S)|^2 \, \mathrm{d}\mu \\ \mathcal{E}_\mu(f;S) &:= \int_S \|\nabla f\|^2 \, \mathrm{d}\mu. \end{aligned}$$

For such a set, the Poincaré constant C_S is defined to be the smallest constant such that

$$\mathrm{Var}_\mu(f;S) \leq C_S \mathcal{E}_\mu(f;S)$$

for all sufficiently regular functions f (noting that it may take the value $+\infty$). A small value of C_S corresponds to a well-connected set S and well-concentrated measure μ, whereas a large value of C_S corresponds to a set S which is poorly-connected, a measure μ which is rather diffuse, or some combination of the two. (Note in passing that the lack of normalisation does not change the value of the optimal constant, i.e. we could (at the expense of additional notation) work with true probability measures throughout.)

Let $A, B \subseteq \mathbf{R}^d$ be measurable and connected sets such that $\mu(A) > 0$, $\mu(B) > 0$, and $\mu(A \cap B) > 0$. Define then ϵ_A, ϵ_B (both in $[0,1]$) by

$$\mu(A \cap B) = (1 - \epsilon_A) \cdot \mu(A) = (1 - \epsilon_B) \cdot \mu(B),$$

noting that both constants are small when A and B overlap substantially.

Context and Contribution

The purpose of this note is to observe that if the Poincaré constant of μ restricted to any two overlapping sets A, B is not too large, then the Poincaré constant of μ restricted to their intersection is not too large either. This result is already known (see e.g. Section 4.6.3, Proposition 4.6.4 of [1], who themselves point to Proposition V.4.4.1 of [2]) in the form of the estimate

$$C_{A \cup B} \leq \frac{\mu(A \cup B)}{\mu(A \cap B)} (C_A + C_B).$$

Actually, the former authors additionally claim that

> "This bound is actually not optimal, but the optimal bound has a less pleasant form".

In any case, this note will provide an improved bound which systematically improves upon both factors, and is perhaps not so unpleasant.

A Basic Lemma

Before proceeding to the main result, we first present a technical lemma.

Lemma 1. *Let $\mathbf{R}^d \supseteq U \supseteq V$ be measurable with $\mu(V) > 0$, and let f be sufficiently regular. It then holds that*

$$|m_\mu(f;U) - m_\mu(f;V)|^2 \leq \left(\frac{1}{\mu(V)} - \frac{1}{\mu(U)}\right)(\mathrm{Var}_\mu(f;U) - \mathrm{Var}_\mu(f;V)).$$

Proof. Using the Cauchy-Schwarz inequality, see that

$$\int_{U\setminus V} f^2 \,\mathrm{d}\mu \geq \frac{1}{\mu(U\setminus V)}\left(\int_{U\setminus V} f\,\mathrm{d}\mu\right)^2 = \frac{(\mu(U)\,m_\mu(f;U) - \mu(V)\,m_\mu(f;V))^2}{\mu(U)-\mu(V)}.$$

Note then the decompositions

$$\begin{aligned}\mathrm{Var}_\mu(f;U) &= \int_U f^2\,\mathrm{d}\mu - \mu(U)\,m_\mu(f;U)^2\\ &= \int_{U\setminus V} f^2\,\mathrm{d}\mu + \int_V f^2\,\mathrm{d}\mu - \mu(U)\,m_\mu(f;U)^2\end{aligned}$$

and

$$\mathrm{Var}_\mu(f;V) = \int_V f^2\,\mathrm{d}\mu - \mu(V)\,m_\mu(f;V)^2$$

to see that

$$\begin{aligned}\mathrm{Var}(f;U) - \mathrm{Var}_\mu(f;V) &\geq \frac{(\mu(U)\,m_\mu(f;U) - \mu(V)\,m_\mu(f;V))^2}{\mu(U)-\mu(V)} - \mu(U)\,m_\mu(f;U)^2 + \mu(V)\,m_\mu(f;U)^2\\ &= \frac{\mu(U)\,\mu(V)}{\mu(U)-\mu(V)}\,|m_\mu(f;U) - m_\mu(f;V)|^2,\end{aligned}$$

and hence conclude by rearrangement. □

Statement and Proof of the Improved Bound

Theorem 2. *Given the setup above, the Poincaré constant $C_{A\cup B}$ of the union can be controlled as*

$$\begin{aligned}C_{A\cup B} &\leq \frac{C_A + C_B + 2\sqrt{\epsilon_A\epsilon_B C_A C_B}}{1-\epsilon_A\epsilon_B}\\ &= \frac{1}{2}\left\{\frac{\left|\sqrt{C_A}+\sqrt{C_B}\right|^2}{1-\sqrt{\epsilon_A\epsilon_B}} + \frac{\left|\sqrt{C_A}-\sqrt{C_B}\right|^2}{1+\sqrt{\epsilon_A\epsilon_B}}\right\}.\end{aligned}$$

Proof. For brevity, write $(a,b,c) = (\mu(A), \mu(B), \mu(A\cap B))$, and fix a sufficiently regular function $f : \mathcal{X} \to \mathbf{R}$. For any scalar $t \in \mathbf{R}$, we can bound the variance on the union $A \cup B$ by decomposing the domain:

$$\begin{aligned}\mathrm{Var}_\mu(f;A\cup B) &\leq \int_{A\cup B} |f-t|^2\,\mathrm{d}\mu\\ &= \int_A |f-t|^2\,\mathrm{d}\mu + \int_B |f-t|^2\,\mathrm{d}\mu - \int_{A\cap B} |f-t|^2\,\mathrm{d}\mu.\end{aligned}$$

By elementary computations, it holds that $\int_S |f-t|^2\,\mathrm{d}\mu = \mathrm{Var}_\mu(f;S) + \mu(S)\cdot|m_\mu(f;S)-t|^2$. Substituting this with $S = A\cup B$ yields that

$$\begin{aligned}\mathrm{Var}_\mu(f;A\cup B) &\leq \mathrm{Var}_\mu(f;A) + \mathrm{Var}_\mu(f;B) - \mathrm{Var}_\mu(f;A\cap B) + R(t),\\ R(t) &= a\,|m_\mu(f;A)-t|^2 + b\,|m_\mu(f;B)-t|^2 - c\,|m_\mu(f;A\cap B)-t|^2.\end{aligned}$$

The remainder term R can be minimised analytically by taking

$$\begin{aligned}(a+b-c)\,t^\star &= am_\mu(f;A) + bm_\mu(f;B) - cm_\mu(f;A\cap B)\\ (a+b-c)\,R^\star &= a(b-c)(m_\mu(f;A) - m_\mu(f;A\cap B))^2\\ &\quad + b(a-c)(m_\mu(f;B) - m_\mu(f;A\cap B))^2\\ &\quad - 2ab(m_\mu(f;A) - m_\mu(f;A\cap B))(m_\mu(f;B) - m_\mu(f;A\cap B)).\end{aligned}$$

Applying Lemma 1 to $(U,V)=(A,A\cap B)$ and $(U,V)=(B,A\cap B)$ respectively yields that

$$\begin{aligned}
\left|m_\mu(f;A)-m_\mu(f;A\cap B)\right|^2 &\le \left(\frac{1}{\mu(A\cap B)}-\frac{1}{\mu(A)}\right)\left(\mathrm{Var}_\mu(f;A)-\mathrm{Var}_\mu(f;A\cap B)\right)\\
&\le \frac{a-c}{ac}\mathrm{Var}_\mu(f;A)\\
\left|m_\mu(f;B)-m_\mu(f;A\cap B)\right|^2 &\le \left(\frac{1}{\mu(A\cap B)}-\frac{1}{\mu(B)}\right)\left(\mathrm{Var}_\mu(f;B)-\mathrm{Var}_\mu(f;A\cap B)\right)\\
&\le \frac{b-c}{bc}\mathrm{Var}_\mu(f;B),
\end{aligned}$$

and so we bound

$$(a+b-c)R^\star \le \frac{(a-c)(b-c)}{c}\left\{\mathrm{Var}_\mu(f;A)+\mathrm{Var}_\mu(f;B)\right\}+2\frac{\sqrt{ab(a-c)(b-c)}}{c}\sqrt{\mathrm{Var}_\mu(f;A)\,\mathrm{Var}_\mu(f;B)}.$$

Granted the algebraic simplifications

$$\begin{aligned}
1+\frac{(a-c)(b-c)}{(a+b-c)c} &= \frac{ab}{(a+b-c)c}=\frac{1}{1-\epsilon_A\epsilon_B},\\
\frac{\sqrt{ab(a-c)(b-c)}}{c(a+b-c)} &= \frac{\sqrt{\epsilon_A\epsilon_B}}{1-\epsilon_A\epsilon_B}
\end{aligned}$$

and dropping the term $\mathrm{Var}_\mu(f;A\cap B)\ge 0$, it follows that

$$\mathrm{Var}_\mu(f;A\cup B)\le \frac{1}{1-\epsilon_A\epsilon_B}\left\{\mathrm{Var}_\mu(f;A)+\mathrm{Var}_\mu(f;B)+2\sqrt{\epsilon_A\epsilon_B}\sqrt{\mathrm{Var}_\mu(f;A)\,\mathrm{Var}_\mu(f;B)}\right\}.$$

Applying the local Poincaré inequalities $\mathrm{Var}_\mu(f;A)\le C_A\mathcal{E}_\mu(f;A)$ and $\mathrm{Var}_\mu(f;B)\le C_B\mathcal{E}_\mu(f;B)$ and observing that the local energies are each bounded by the total energy $\mathcal{E}_\mu(f;A\cup B)$ (as each energy integrates $\|\nabla f\|^2\ge 0$ over a subset of $A\cup B$), we obtain that

$$\mathrm{Var}_\mu(f;A\cup B)\le \frac{C_A+C_B+2\sqrt{\epsilon_A\epsilon_B C_A C_B}}{1-\epsilon_A\epsilon_B}\mathcal{E}_\mu(f;A\cup B)$$

Recalling that f was arbitrary, we conclude that

$$C_{A\cup B}\le \frac{C_A+C_B+2\sqrt{\epsilon_A\epsilon_B C_A C_B}}{1-\epsilon_A\epsilon_B}$$

as claimed. The alternative expression follows from the algebraic identity $x^2+2\alpha xy+y^2=\frac{1+\alpha}{2}(x+y)^2+\frac{1-\alpha}{2}(x-y)^2$. □

Remark 3. Revisit for a minute the bound presented in [2, 1]

$$C_{A\cup B}\le \frac{\mu(A\cup B)}{\mu(A\cap B)}(C_A+C_B),$$

which in our notation, might be written as

$$\bar{C}^{\mathrm{Fr}}_{A\cup B}\le \frac{1-\epsilon_A\epsilon_B}{(1-\epsilon_A)(1-\epsilon_B)}(C_A+C_B).$$

To see the improvement of our new bound, it is instructive to consider the case in which $(\mu(A),\mu(B),C_A,C_B)$ all remain of constant order as the overlap c tends to 0, whereupon the bounds behave as

$$\begin{aligned}
c\bar{C}^{\mathrm{Fr}}_{A\cup B} &\to (a+b)(C_A+C_B)\\
&= 4\left(\frac{\mu(A)+\mu(B)}{2}\right)\left(\frac{C_A+C_B}{2}\right)\\
c\bar{C}^{\mathrm{SP}}_{A\cup B} &\to \frac{ab}{a+b}\left(C_A+C_B+2\sqrt{C_A C_B}\right)\\
&= 2\left(\frac{\mu(A)^{-1}+\mu(B)^{-1}}{2}\right)^{-1}\left(\frac{\sqrt{C_A}+\sqrt{C_B}}{2}\right)^2.
\end{aligned}$$

Observe that by Jensen's inequality, each of the two factors is improved by the new bound; moreover, the new bound improves unilaterally by a factor of 2.

Extensions and Further Remarks

(1) These calculations scarcely use the interpretation of the Poincaré constant as the relaxation time of the Overdamped Langevin dynamics, and indeed, the same calculations furnish bounds on the spectral gap of more-or-less arbitrary reversible Markov kernels, being careful to appropriately re-define e.g.

$$\mathcal{E}_{\mu,P}\left(f;S\right) := \int_{S\times S} \mu\left(\mathrm{d}x\right)\ P\left(x,\mathrm{d}y\right) \frac{\left[f\left(x\right)-f\left(y\right)\right]^2}{2},$$

and so on. An interesting refinement would be to seek analogous results for 'weak' Poincaré inequalities using the same machinery.

(2) It stands to reason that the Poincaré constant is not all that special here, and that one could find analogous controls of other qualitatively similar quantities, e.g. the Cheeger constant or the Log-Sobolev constant. Preliminary explorations suggest that while the overall proof structure largely works in these two cases,
   (a) for the Cheeger constant, the step of optimising over the level t becomes difficult (since minimising sums of terms of the form $t \mapsto |f-t|$ is not quite so tractable as in the quadratic case), and this leads to somewhat looser bounds, whereas
   (b) for the Log-Sobolev constant, the optimisation over the level t remains tractable (essentially due to the entropy's structure as a Bregman divergence), and the challenge is instead to adapt a suitable 'entropic' variant of our Lemma 1.

Acknowledgments

I first recall encountering the original bound of [2, 1] in late 2023, where it was made clear in the exposition that the bound was clean but marginally loose. At the time, I spent about an afternoon trying to sharpen the bound (following the original proof structure given in [1]), but these attempts were ultimately unsuccessful (minimising a trivariate quadratic by hand turned out to be a bit too much for me, I have to infer).

Today (in April 2026), I was cleaning up old tweets, which reminded me of this problem. Having had some success with problems of a similar type, I felt encouraged to revisit this 'union bound' with the help of mathematical tools which have advanced somewhat in the intervening years. As such, I typed up the problem in LyX as clearly as I could, passed the question to Gemini 3.1 Pro Preview. After a few minutes' wait, this then supplied more-or-less the proof above. I cleaned up the presentation (e.g. isolating Lemma 1 as its own object, re-labelling the proportions ϵ_A, ϵ_B, providing the alternate expression for $\bar{C}^{\mathrm{SP}}_{A\cup B}$), incorporated some additional remarks, and finalised the draft.

References

[1] Dominique Bakry, Ivan Gentil, and Michel Ledoux. *Analysis and geometry of Markov diffusion operators*, volume 103. Springer, 2014.

[2] Jacques Franchi and Yves Le Jan. *Hyperbolic dynamics and Brownian motion: an introduction.* Oxford Mathematical Monographs, 2012.